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Afatinib versus cisplatin plus gemcitabine for first-line treatment of Asian patients with advanced non-small-cell lung cancer harbouring EGFR mutations (LUX-Lung 6): an open-label, randomised phase 3 trial

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Abstract

Background: Afatinib—an oral irreversible ErbB family blocker—improves progression-free survival compared with pemetrexed and cisplatin for first-line treatment of patients with EGFR mutation-positive advanced non-small-cell lung cancer (NSCLC). We compared afatinib with gemcitabine and cisplatin—a chemotherapy regimen widely used in Asia—for first-line treatment of Asian patients with EGFR mutation-positive advanced NSCLC.

Methods: This open-label, randomised phase 3 trial was done at 36 centres in China, Thailand, and South Korea. After central testing for EGFR mutations, treatment-naïve patients (stage IIIB or IV cancer [American Joint Committee on Cancer version 6], performance status 0–1) were randomly assigned (2:1) to receive either oral afatinib (40 mg per day) or intravenous gemcitabine 1000 mg/m² on day 1 and day 8 plus cisplatin 75 mg/m² on day 1 of a 3-week schedule for up to six cycles. Randomisation was done centrally with a random number-generating system and an interactive internet and voice-response system. Randomisation was stratified by EGFR mutation (Leu858Arg, exon 19 deletions, or other; block size three). Clinicians and patients were not masked to treatment assignment, but the independent central imaging review group were. Treatment continued until disease progression, intolerable toxic effects, or withdrawal of consent. The primary endpoint was progression-free survival assessed by independent central review (intention-to-treat population). This study is registered with ClinicalTrials.gov, NCT01121393.

Findings: 910 patients were screened and 364 were randomly assigned (242 to afatinib, 122 to gemcitabine and cisplatin). Median progression-free survival was significantly longer in the afatinib group (11·0 months, 95% CI 9·7–13·7) than in the gemcitabine and cisplatin group (5·6 months, 5·1–6·7; hazard ratio 0·28, 95% CI 0·20–0·39; p<0·0001). The most common treatment-related grade 3 or 4 adverse events in the afatinib group were rash or acne (35 [14·6%] of 239 patients), diarrhoea (13 [5·4%]), and stomatitis or mucositis (13 [5·4%]), compared with neutropenia (30 [26·5%] of 113 patients), vomiting (22 [19·5%]), and leucopenia (17 [15·0%]) in the gemcitabine and cisplatin group. Treatment-related serious adverse events occurred in 15 (6·3%) patients in the afatinib group and nine (8·0%) patients in the gemcitabine and cisplatin group.

Interpretation: First-line afatinib significantly improves progression-free survival with a tolerable and manageable safety profile in Asian patients with EGFR mutation-positive advanced lung NSCLC. Afatinib should be considered as a first-line treatment option for this patient population.

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[Afatinib in NSCLC harbouring EGFR mutations.](#)

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